

CARBON COUNTY REGIONAL, UT, USA

WMO: 724700

Lat: 39.609N

Lon: 110.755W

Elev: 5921

StdP: 11.81

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 93141

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.1	9.5	-6.6	4.8	16.8	-2.5	6.0	20.6	22.9	32.2	20.0	30.2	3.7	10	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.9	93.6	58.2	91.3	57.6	89.3	57.2	62.6	81.9	61.5	81.1	60.3	80.1	10.4	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.3	87.6	66.4	55.1	80.7	66.9	53.0	74.7	67.8	31.2	81.9	30.2	81.2	29.3	79.9	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.2	19.1	16.9	-0.3	98.6	4.4	4.0	-3.5	101.4	-6.1	103.8	-8.6	106.0	-11.8	108.9
			-1.3	66.0	4.3	3.6	-4.4	68.6	-6.9	70.7	-9.3	72.7	-12.4	75.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	26.3	31.3	41.1	47.8	57.0	68.4	75.4	72.1	63.7	50.0	37.2	26.5	
	DBStd	18.30	7.69	7.61	7.24	7.23	7.92	7.07	4.47	4.31	6.61	7.23	7.76	7.58	
	HDD50	2903	734	523	286	126	28	1	0	0	4	86	387	728	
	HDD65	6305	1199	943	740	516	265	47	2	4	96	465	835	1193	
	CDD50	2850	0	0	11	59	246	554	788	686	416	88	2	0	
	CDD65	777	0	0	0	0	18	150	325	225	58	1	0	0	
	CDH74	10203	0	0	0	20	403	2290	3962	2608	887	33	0	0	
	CDH80	4384	0	0	0	2	94	993	1950	1100	242	3	0	0	
Wind	WSAvg	6.7	4.9	5.8	7.6	8.6	8.1	7.9	6.9	6.6	6.7	6.5	6.0	5.0	
Precipitation	PrecAvg	12.4	1.0	1.1	1.0	1.0	1.1	0.7	0.9	1.4	1.5	1.3	0.7	1.0	
	PrecMax	15.6	2.6	2.7	2.4	2.5	3.0	1.9	1.9	3.3	3.7	3.7	1.9	2.5	
	PrecMin	8.7	0.0	0.0	0.0	0.2	0.1	0.1	0.2	0.3	0.2	0.0	0.0	0.1	
	PrecStd	1.9	0.8	0.7	0.6	0.6	0.7	0.5	0.4	0.7	1.0	0.9	0.4	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.9	57.4	70.2	76.6	87.6	95.4	97.4	94.2	89.2	78.7	62.8	50.2	
		MCWB	37.0	40.6	45.8	48.9	57.0	57.6	60.2	58.7	56.4	51.5	44.5	38.2	
	2%	DB	45.0	52.0	64.2	71.9	82.2	92.3	94.8	91.4	85.7	72.3	58.8	45.4	
		MCWB	35.4	38.5	43.6	47.1	53.0	56.6	58.8	57.8	55.0	49.4	42.5	35.3	
	5%	DB	41.3	47.3	60.2	68.1	78.9	89.6	92.4	89.5	82.6	67.9	54.9	41.6	
		MCWB	32.1	36.5	41.8	45.6	51.0	55.9	58.4	57.4	53.9	47.3	40.7	33.3	
	10%	DB	37.6	43.2	55.3	63.9	74.6	86.2	90.2	86.7	79.8	64.0	50.4	38.0	
		MCWB	30.9	34.3	39.9	43.9	49.7	54.6	57.9	57.0	53.1	45.9	38.5	31.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	42.1	47.1	50.6	58.8	62.0	65.4	64.1	60.5	55.1	46.8	40.1
			MCDB	46.0	53.9	66.2	72.2	85.3	89.6	86.0	82.3	74.5	67.6	59.9	46.5
2%		WB	36.3	39.6	44.6	48.4	54.6	59.4	63.3	62.4	58.7	51.8	43.8	37.0	
		MCDB	42.1	49.4	61.7	68.3	77.1	84.5	82.8	80.0	74.6	65.6	55.4	42.7	
5%		WB	34.0	37.4	42.6	46.5	52.4	57.6	61.9	61.3	57.0	50.1	41.6	34.3	
		MCDB	39.6	44.9	57.7	64.8	74.2	82.4	82.3	79.1	74.6	63.6	53.3	40.7	
10%		WB	32.0	35.4	40.7	44.8	50.7	56.0	60.8	60.0	55.4	47.9	39.4	32.0	
		MCDB	37.1	41.9	54.2	62.0	71.6	81.0	81.6	77.9	73.2	61.4	49.2	37.7	
Mean Daily Temperature Range		5% DB	MDBR	18.5	18.9	22.5	24.9	25.9	29.2	27.9	27.0	26.7	23.2	20.8	18.0
			MCDBR	21.5	23.8	28.8	31.1	31.9	33.3	31.0	31.0	30.9	28.0	25.5	21.6
	MCWBR		14.0	13.7	14.4	14.8	13.8	14.3	12.2	12.2	13.1	13.5	14.0	13.8	
	5% WB	MCDBR	18.8	20.4	26.5	28.4	28.9	29.6	25.8	24.9	24.7	23.2	23.3	19.3	
		MCWBR	12.6	12.5	13.8	14.0	12.9	13.7	10.6	10.3	11.1	11.4	13.5	13.0	
Clear-Sky Solar Irradiance	taub	0.241	0.253	0.267	0.300	0.304	0.307	0.347	0.345	0.299	0.279	0.250	0.244		
	taud	2.512	2.525	2.534	2.440	2.492	2.503	2.421	2.428	2.552	2.572	2.593	2.541		
	Ebn at Noon	303	313	318	310	309	306	293	291	302	299	298	294		
	Edh at Noon	23	27	30	35	34	34	36	35	29	25	21	21		
All-Sky Solar Radiation	RadAvg	786	1062	1522	1894	2176	2489	2251	1986	1739	1268	889	672		
	RadStd	64	98	102	101	149	174	119	101	106	105	65	58		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page